

Go Daddy Root Certificate Authority - G2	
Go Daddy Secure Certificate Authority - G2	
*.tutorialspoint.com	
Details	
Subject Name	
Common Name	*.tutorialspoint.com
Issuer Name	
Country or Region	US
County	Arizona
Locality	Scottsdale
Organisation	GoDaddy.com, Inc.
Organisational Unit	http://certs.godaddy.com/repository/
Common Name	Go Daddy Secure Certificate Authority - G2
Serial Number	00 D8 0C E2 2E 2B B5 7B 80
Version	3
Signature Algorithm	SHA-256 with RSA Encryption (1.2.840.113549.1.1.11)
Parameters	None
Not Valid Before	Wednesday, 2 February 2022 at 11:11:36 India Standard Time
Not Valid After	Wednesday, 15 February 2023 at 14:56:00 India Standard Time
Public Key Info	
Algorithm	RSA Encryption (1.2.840.113549.1.1.1)
Parameters	None
Public Key	256 bytes: C5 56 30 01 06 3E 9F CD ...
Exponent	65537
Key Size	2,048 bits
Key Usage	Encrypt, Verify, Wrap, Derive
Signature	256 bytes: 89 19 B0 36 1C 6F 97 41 ...
Extension	Key Usage (2.5.29.15)
Critical	YES
Usage	Digital Signature, Key Encipherment
Extension	Basic Constraints (2.5.29.19)
Critical	YES
Certificate Authority	NO
Extension	Extended Key Usage (2.5.29.37)
Critical	NO

Figure 4: Example of certificate attributes

Organisation validation (OV)

certificate: Provides stronger assurance than the DV; organisation verification is done manually and the certificate takes time to be issued.

Extended validation (EV)

certificate: Uses a green padlock to indicate that the owner has been verified. Rigorous checks are performed on the subject. EV certificates can't have wildcard domains, and provide assurance against phishing attacks. They also take some time to be issued after validation.

Server certificate: Part of the web server certificate, it guarantees the identity of the user.

Machine computer certificate:

This is issued to machines/servers/routers/switches. Machines without a valid certificate are not allowed in the network.

Email certificate: This is issued to decrypt and encrypt messages, and is used by secure layer protocols such as S/MIME, POPv3S, PGP, IMAPS, and so on.

Code signing certificate: Software

publishers use this certificate to verify source code.

Root certificate: This is used to identify the certificate's root. Usually, the key size ranges from 2048 bytes to 4096 bytes, and more.

Self-signed certificate: Users create this for themselves, but it's not trusted and generally used only for testing purposes.

User certificate: This identifies individuals. Also known as client certificates, it works like how passwords work.

Certificate formats

- DER - Binary format of the certificate.
- PEM - Binary data represented in ASCII text format using Base64 encoding.
- CER - Used in Windows; file extension is *.cer. Not a standard form.
- CRT - Used in Linux; file extension is *.crt. Not a standard form.
- PKCS - The contents of the file with this extension hold more than one certificate. Usually, these files are and need to be password protected. In Windows this is created with .PFX extension and in Mac or iOS with .P12 extension. In Linux, keys are stored in separate files. This file is used to transfer keys or for backup.
- P7B - This is similar to PKCS but stores information in the ASCII format and doesn't contain a primary key. Usually used in chain of trust or secure email.
- KEY - Used only when the file holds the primary key.

Key recovery

Private keys should be shelved with the highest security and minimal access controls.

Key recovery defines a secure way to save backups, and to retrieve a key when lost.